

Architecture Selection

I chose Microservices Architecture for the Clinical Trial Subject Recruitment Engine.

Reason 1 – Independent scalability:

The eligibility screening engine must process up to 500 concurrent survey submissions with a 95th-percentile response time below 800 ms. Separating eligibility evaluation into an independent service allows the screening functionality to be scaled independently when recruitment traffic increases.

Reason 2 – Independent functionality:

The system contains separate functions such as eligibility evaluation, protocol criteria management, electronic consent, subject management, and notifications. Microservices allow these functions to be independently deployed and maintained while communicating through defined interfaces. The architecture therefore supports changes to trial criteria without requiring changes to the entire system.

Security Advantage:

A dedicated Audit & Compliance Service can centralize immutable audit logging for subject-data operations. This supports the requirement for 21 CFR Part 11 and HIPAA compliance, including cryptographic audit trails and protection of stored PHI.

Performance Benefit:

The eligibility evaluation functionality can be independently scaled during high-volume recruitment campaigns, helping the system meet its sub-second screening requirement and 99.9% availability target.